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Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Artificial Intelligence (AI)	Topic	Computer Vision & Recommendation Systems	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

- Read each question carefully before answering.
- For Section A and Section B, mark one answer only.
- For Sections C and E, write your answer clearly in the space provided.
- For Section D, write the correct letter from Column B against each item in Column A.
- Do not use correction fluid. Keep all answers clear and legible.
- This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. What is computer vision?

A. A type of computer monitor

B. The field of AI that enables computers to interpret and understand visual information

C. A brand of camera lens

D. Software for editing spreadsheets

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. What is "object detection" in computer vision?

A. Deleting unwanted files from a folder

B. Repairing a broken camera

C. Identifying and locating specific objects within an image

D. Compressing an image file

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. How does computer vision typically analyse an image?

A. By examining pixels and learned patterns to recognise features

B. By printing the image and reading it manually

C. By asking a human to describe it every time

D. By ignoring the image content entirely

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. What is a convolutional neural network (CNN) commonly used for?

A. Sending emails

B. Processing and analysing image data

C. Formatting text documents

D. Managing a computer's power supply

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. Which of these is a real-world application of computer vision?

A. A basic text editor

B. A plain calculator app

C. A USB flash drive

D. A self-driving car recognising road signs and pedestrians

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. What is facial recognition? A. A method of encrypting files C. Identifying a person's identity from an image of their face Answer: ☐ A ☐ B ☐ C ☐ D	B. A way to compress video D. A type of computer password Intermediate · 2 pts
7. What can make computer vision less accurate? A. Poor lighting or unusual camera angles C. Having too much storage space Answer: ☐ A ☐ B ☐ C ☐ D	B. Using a faster internet connection D. Using a newer keyboard Intermediate · 2 pts
8. What is a recommendation system? A. A system that repairs broken software C. A type of computer virus Answer: ☐ A ☐ B ☐ C ☐ D	B. A system that suggests content or products based on user data and preferences D. A tool for managing passwords Intermediate · 2 pts
9. What do recommendation systems typically base their suggestions on? A. Random selection with no data involved C. A user's past behaviour, preferences, and activity Answer: ☐ A ☐ B ☐ C ☐ D	B. The weather forecast D. The device's battery level Intermediate · 2 pts
10. What is "collaborative filtering"? A. Recommending items based on what similar users liked C. Sorting files by date only Answer: ☐ A ☐ B ☐ C ☐ D	B. Deleting duplicate files D. Blocking spam emails Intermediate · 2 pts
11. What is "content-based filtering"? A. Blocking inappropriate content automatically C. Filtering out low-quality images Answer: ☐ A ☐ B ☐ C ☐ D	B. Recommending items similar to what a user has liked before D. Removing viruses from a file Intermediate · 2 pts
12. Which platforms commonly use recommendation systems? A. A basic calculator app C. A basic alarm clock app Answer: ☐ A ☐ B ☐ C ☐ D	B. A plain text file viewer D. Streaming services like Netflix and Spotify Advanced · 2 pts
13. Which type of data do recommendation systems typically use? A. Watch history, ratings, and past purchases C. Only the device's screen brightness Answer: ☐ A ☐ B ☐ C ☐ D	B. Only the current date and time D. The colour of the app icon Advanced · 2 pts
14. What is the main purpose of a recommendation system? A. To slow down the user's browsing C. To delete a user's account Answer: ☐ A ☐ B ☐ C ☐ D	B. To personalise suggestions and help users discover relevant content D. To increase a device's battery life Advanced · 2 pts
15. Which of these best describes how computer vision and recommendation systems are similar? A. They are exactly the same technology C. Both are AI applications that learn patterns from data to make useful predictions Answer: ☐ A ☐ B ☐ C ☐ D	B. Neither uses any data at all D. Both only work without an internet connection Advanced · 2 pts

Section B — True / False

20 marks

16. Computer vision enables computers to interpret and understand visual information from images and video. A. True ☐ B. False ☐	Basic · 2 pts
17. Computer vision cannot be used in self-driving cars. A. True ☐ B. False ☐	Basic · 2 pts
18. Convolutional neural networks (CNNs) are commonly used for processing image data. A. True ☐ B. False ☐	Basic · 2 pts
19. Facial recognition is completely unrelated to computer vision. A. True ☐ B. False ☐	Basic · 2 pts
20. Recommendation systems suggest content based on a user's data and behaviour. A. True ☐ B. False ☐	Intermediate · 2 pts

21. Collaborative filtering recommends items based on what similar users liked.
A. True ☐ B. False ☐

22. Content-based filtering completely ignores what a user has liked in the past.
A. True ☐ B. False ☐

23. Netflix and Spotify both use recommendation systems to personalise suggestions.
A. True ☐ B. False ☐

24. Lighting and camera angle can affect the accuracy of computer vision systems.
A. True ☐ B. False ☐

25. Recommendation systems never use a user's past behaviour or history.
A. True ☐ B. False ☐

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Section C — Fill in the Blank

20 marks

26. _____ vision is the field of AI that enables computers to interpret and understand visual information.
Answer: _____

27. _____ detection identifies and locates specific objects within an image.
Answer: _____

28. _____ recognition identifies a person's identity from an image of their face.
Answer: _____

29. A _____ neural network (CNN) is commonly used to process image data.
Answer: _____

30. A _____ system suggests content or products based on user data and preferences.
Answer: _____

31. _____ filtering recommends items based on what similar users liked.
Answer: _____

32. _____-based filtering recommends items similar to what a user has liked before.
Answer: _____

33. Recommendation systems analyse past _____ such as watch history and ratings.
Answer: _____

34. Streaming platforms such as Netflix use recommendation systems to suggest _____.
Answer: _____

35. Factors such as lighting and camera _____ can affect computer vision accuracy.
Answer: _____

Basic · 2 pts

Basic · 2 pts

Basic · 2 pts

Basic · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Section D — Column Matching

10 marks

36-40. Match each term to its correct definition.

Column A	Answer	Column B
Computer Vision	_____	A. Identifying and locating specific objects within an image
Object Detection	_____	B. Identifying a person's identity from an image of their face
Facial Recognition	_____	C. Recommending items based on what similar users liked
Collaborative Filtering	_____	D. Recommending items similar to what a user has liked before
Content-Based Filtering	_____	E. The field of AI that enables computers to interpret visual information from images and video

Section E — Short Answer

20 marks

41. What is computer vision?

Advanced · 4 pts

42. Give an example of a real-world application of computer vision.

Advanced · 4 pts

43. What is the key difference between collaborative and content-based filtering?

Advanced · 4 pts

44. Give an example of a platform that uses a recommendation system.

Advanced · 4 pts

45. What kind of data do recommendation systems typically use?

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

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Computer Vision & Recommendation Systems — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. B | 9. C |
| 2. C | 10. A |
| 3. A | 11. B |
| 4. B | 12. D |
| 5. D | 13. A |
| 6. C | 14. B |
| 7. A | 15. C |
| 8. B | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. False | 24. True |
| 20. True | 25. False |

Section C — Fill in the Blank

- | | |
|--------------------|-------------------|
| 26. computer | 31. collaborative |
| 27. object | 32. content |
| 28. facial | 33. behaviour |
| 29. convolutional | 34. content |
| 30. recommendation | 35. angle |

Section D — Column Matching

- | | |
|----------------------------|---------------------------------|
| 36. Computer Vision — E | 39. Collaborative Filtering — C |
| 37. Object Detection — A | 40. Content-Based Filtering — D |
| 38. Facial Recognition — B | |

Section E — Short Answer (Model Answers)

- 41.** image; visual; interpret; understand; video
- 42.** self-driving; medical; security; face; camera; photo; car
- 43.** similar users; similar items; other users; past likes; user; item
- 44.** netflix; spotify; amazon; youtube
- 45.** history; rating; click; behaviour; behavior; purchase; watch